# AI Powered Agriculture Optimization Chatbot using RAG and GenAI

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***Abstract* — Agriculture plays an important role in economic development as well as food security around the world, though farmers face a big problem of lack of access to compliance information that can help them to improve their Agriculture. In this paper, we propose a chatbot for farmers to pose their questions and get the most appropriate and extensive answers instantly with the help of retrieval augmented generation technique. The chatbot incorporates artificial intelligence, on-demand service provision, and access to farmers' requirements and occurrences from written literature. Chatbots are a versatile tool that can assist and establish a procedure with the purpose of delivering effective, engaging and helpful information. This results into better decisions by the farmers and improved returns on the farming units. From the assessment of the application, it is possible to see the advantages of the proposed system for agricultural production using the permaculture concept and raising awareness among farmers. This study examines the development, implementation and operation of the chatbot, demonstrating its potential to transform agricultural advisory services and improve smart agriculture. Together, the chatbot is unlocking the transformative potential of AI to further transform agricultural services, promote permaculture and help farmers make informed decisions.**

***Keywords — Agriculture, Chatbot, GenAI, NLP, Retrieval-Augmented Generation, Smart Farming.***

## I. INTRODUCTION

The agricultural environment has undertaken significant changes, Driven by technological developments, especially in precision agriculture. From [8], This data-driven approach promises to improve crop quality, resource allocation and overall agricultural management. The key to the success of precision agriculture is the provision and dissemination of agricultural information to farmers, enabling them to make informed decisions about various agricultural practices.

There are numerous challenges which occur in farming globally; these include climatic conditions, market volatility, and many others; there is high need of improved facility in helping in decision-making within the agricultural sector. Through this chatbot, the revolution is being availed not only in making all the information that the farmers may need regarding farming be made available to them, but even with the options that may be tailored to suit the needs of every farmer including him. As per [11], Comparing to crop management, pest control, soil fertility and market prediction, the proposed system is eagerly waiting to increase farm yields and yield per unit area by applying sustainable agriculture technologies. This being, true with AI updates constantly emerging, uses of this system will look like the following: Hence, the proposed system will become a real tool to change Agriculture for better, and support the food systems of the globe.

Traditional methods of knowledge dissemination, such as agricultural extension services and physical manuals, often lack scalability, reach, and accessibility. To address these limitations, from [12] AI-powered chatbots have emerged as a compelling solution. Leveraging AI algorithms, chatbots provide farmers with instant access to a wealth of agricultural information in a conversational and user-friendly manner, revolutionizing the interaction with agricultural knowledge.

In this paper, we introduced an AI-powered chatbot meticulously designed to meet the information needs of farmers. Through this, the Retrieval-Augmented Generation model, the system adapts the Retrieval based and the Generation based methods to adequately get farmers thorough answers to their questions. The rest of the paper is structured as follows: Literature Review is discussed in Section II. In Section III, details about the methodology including the system architecture, RAG, LLM & MiniLM transformers are presented. Section IV discusses result analysis. Finally, the paper is concluded in Section V.

## II. LITERATURE REVIEW

In research paper [1], the authors describe the use of an intelligent chatbot to tackle the problems encountered in farm productions. They created the chatbot using RASA X that possesses RASA NLU and RASA Core. In response to weather inquiries, they integrated the Open Weather Map API to ensure that the consumers received the current climate information. For the implementation of the chatbot, the application was hosted on Twilio and ngROK on WhatsApp. In particular, the accuracy of the chatbot was 96%. 5%, in correctly diagnosing positive cases on the other hand was 1%.

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In research paper [2], the authors also trained and tested two machine learning models, which includes Gaussian Naive Bayes and Support Vector Machine that are useful in recommending crops to farmers, also trained the VGG-16 deep learning model for the identification of crop diseases through image analysis. The research paper showed that such models can indeed help in enhancing the practices of agriculture and help farmers in decision-making.

According to the study in paper [3], the authors concern the advanced NLP models for technical university information chatbots. According to them, models based on the neural network like Hercules, which uses the sequential model to prevent overfitting, are better by utilizing accuracy and handling different queries and with spelling mistakes. Lastly, the paper provides a discussion on the time complexity comparison of various chatbot solutions, which outlines the capability of the proposed approaches.

This research paper [4] presents an integrated approach for enhancing crop productivity by addressing three key aspects: crop disease prediction, crop recommendation, and crop yield prediction. The researchers used machine learning algorithms such as decision trees, random forests, and convolutional neural networks to develop a predictive model for early detection of crop diseases. The researchers employed a personalized approach using machine learning techniques like k-nearest neighbors, collaborative filtering, and matrix factorization to provide suitable crop recommendations based on environmental conditions and user preferences, also used data-driven approaches, including regression techniques and time-series analysis, to develop a model that can accurately forecast future crop yields.

The [5] paper discusses the use of VGG19 technique for image feature extraction and recognition to detect 24 different crop diseases, the use of geometric (cropping, rotating, shifting) and photometric (brightness) transformation techniques to augment the training data, reviews the critical challenges and future trends in developing flexible, open-source, and AI-based smart agriculture systems. Thus, the paper proposes an architecture for an AI-based fertilizer recommendation system that can sense agricultural parameters, process data, and provide recommendations to farmers.

This research paper [6], the author presents a case study on developing an artificial intelligence-based chatbot model to provide expert advice to potato farmers in Kenya. The paper outlines a challenge of limited access to an expert and certified potato seed producers availed to Kenyan potato farmers, which has, in turn, caused low potato yields it explain the integrated architecture categorizing the proposed AI-based chatbot model, which comprises of IBM Watson Assistant, Geographic Information System, and OutSystems to offer expert advice and connect potato farmers with certified seed producers.

The research paper [7], discusses the development of a chatbot system that leverages domain-specific artificial intelligence to enhance virtual assistance. The authors aim to create a chatbot that goes beyond general-purpose AI by integrating specialized knowledge and capabilities personalized to specific domains, making the interaction more relevant and effective for users.

## III. METHODOLOGY

This research proposes AI-powered chatbot as shown in Fig. 1, which is designed to empower farmers with information accessibility. The chatbot leverages a Retrieval-Augmented Generation model to address farmers' agricultural queries. The RAG model retrieves the most relevant information from a comprehensive repository of agricultural documents and utilizes that information to generate informative and comprehensive responses tailored to each user's specific question. To gain a deeper understanding of its functionality, we will explore the following aspects in detail: system architecture, data preparation techniques, RAG model implementation, and the evaluation strategy employed to assess its effectiveness in assisting farmers.

**RAG:** In [10], Retrieval-augmented generative models can be considered as an invention that aims at examining both the retrieval and generation concepts together with other models as dual systems. It has an access component, namely able to search for relevant data in the knowledge base depending on the data provided by a user and a rendering component capable of using the summarized content

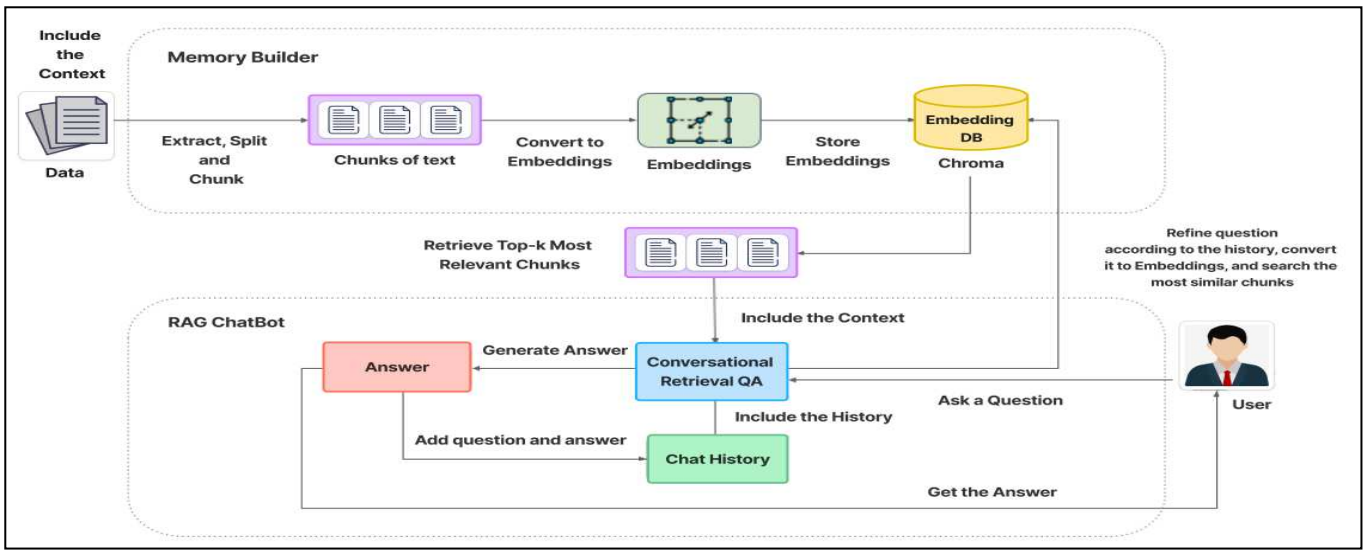


Fig. 1. System Architecture

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acquired from the data base to come up with a natural language response in a question & answer format that is straightforward and can be easily replied. The benefit of the RAG model is to create a safe and factually accurate response that is idiomatic to real-life questions and concerns across the subject matter. It is particularly useful especially for topics such as Agriculture where it can prove to be very important, especially in making sure that the farmers get the right information. Just aggregating that kind of knowledge of RAG Model to management of crops, diseases, business, permaculture, and many more and it will become very effective in doing all of them. In [9] its general context, it would be reasonable to state that such a RAG model is a next-gen model in NLP area and shows potential with the strength and the flexibility that come with the combination of retrieval and language generation power in order to spit out the right answers. The RAG model employed in this study integrates retrieval mechanisms with generative capabilities, formulated as:

$$P(x|y) = \sum_{d \in D} P(d|x)\, P(z|x, d) \qquad (1)$$

**LLM:** In [14], Large language models are new-age AI tools employed for deciphering big data and understanding relations between segments of a language, including NLP. As applied to the context of the proposed system which is a conversational AI assist for farmers, the use of the LLMs and the RAG model holds several strengths given by the capacity to post questions to huge knowledge repositories by topic and build responses from the inquiry of the farmer. Similarly, from [13] as LLMs like GPT series based on OpenAI aims at learning all the textual data in understanding human language, there are challenges like in the development of this conversational commerce chatbot in the agricultural sector, the LLMs needs to be fine-tuned to suit the agriculture business. Therefore, using such LLMs, with the support of this chatbot, farmers can be offered the necessary information that is relevant, reliable and relevant to the particular situation in order to influence the increase in yields, productivity and efficiency of the agricultural sector. The chatbot uses the Transformer model, which employs a self-attention mechanism defined by:

$$Attention(Q, K, V) = softmax\left(\frac{QK^T}{\sqrt{d_k}}\right)V \qquad (2)$$

**MiniLM Transformer:** To evaluate the effectiveness of different transformer models in our NLP tasks, we included the MiniLM Transformer as one of the primary models in our experiments. MiniLM, known for its efficiency, was selected due to its ability to deliver high performance while minimizing computational overhead. The model was fine-tuned on our specific dataset, with key parameters such as learning rate, batch size, and epoch count carefully optimized. Throughout the training process, MLflow was used to track and log the performance of MiniLM, allowing for a detailed comparison with other models like BERT and RoBERTa.

**Fine Tuning:** In this research work, Fine-tuning is crucial for optimizing the performance of the AI chatbot for farmers, when leveraging advanced language models like the RAG model. The fine tuning includes the use of general pre-trained weights for the RAG model followed by training of the model on agriculture specific datasets that include farming techniques, crop handling, pest eradication and markets. This helps the model to be tuned to the language and the terminologies that farmers use in the course of carrying out their work. It also includes the cycle of tuning hyperparameters and can utilize the transfer learning in the related domains. With proper tuning, the proposed RAG model can be used for determining the most appropriate and correct advice to offer to the farmers based on the context or situation a farmer may be in and thus make this system a more practical and useful chatbot. Fine-tuning of the chatbot was done using the cross-entropy loss function with the purpose of adjusting the model weights for better fit according to the prediction errors, formulated as:

$$L = -\frac{1}{N} \sum_{i+1}^{N} \sum_{j+1}^{C} y_{ij} \log(\hat{y}_{ij}) \qquad (3)$$

### A. *Data Preprocessing*

- **Data Collection:** As shown in Fig. 2, The first step includes determining the kind of data that shall be retrieved in the building of the chatbot's knowledge base and this include websites data, documents, articles or any other textual information.
- **Data Cleaning and Preprocessing:** Raw data obtained from various sources is then cleansed to minimize the inclusion of the noise or unwanted data. This process might involve activities like dropping the stop words,

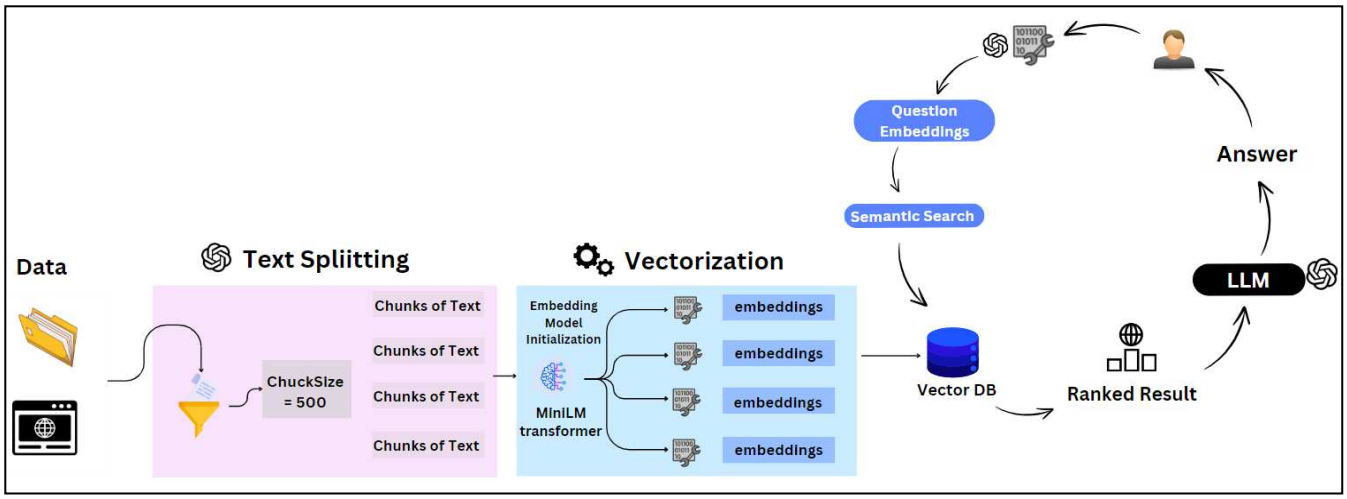


Fig. 2. Process Flow

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stemming or lemmatization of terms, and formatting and encoding the properties if necessary.

- **Chunking and Indexing:** The preprocessed data is portioned into smaller portions that are easier to process (e.g., the paragraph or sentence). These chunks are then indexed and stored in a database or any other structure convenient for identification the next time they are needed.

*B. Embedding Generation*

- **Embedding Conversion:** As shown in Fig. 2, These chunks of the text are then converted into vectors or features commonly referred to as embeddings with the help of an embedding function or a language model like BERT or GPT. Such embeddings store the information relating to the meaning and context of the text for efficient search and analysis.
- **Embedding Storage:** The generated embeddings are stored in a dedicated Embedding Database, which serves as the knowledge base for the chatbot.

*C. Retrieval and Generation*

- **Query Refinement:** As shown in Fig. 2, While submitting a question, the system cleans up a query in a manner to filter it by using conversation history to better understand intent and context of the user.
- **Retrieval of Relevant Chunks:** The derived fine query is used to find out the chunk that is semantically most similar to the fine form of the query present in Embedding Database by identifying the closest matches.
- **Context-Aware Generation:** Therefore, the chunks are mixed with the conversation history and then passed into a generation model, for example, a language model that has been pre-trained and fine-tuned for the task, to generate a reasonable and related response to a question asked by the user.
- **Response Delivery:** The generated response is then displayed to the user and this completes the conversational interaction.

## IV. Result Analysis

### 1) Analysis of User Queries

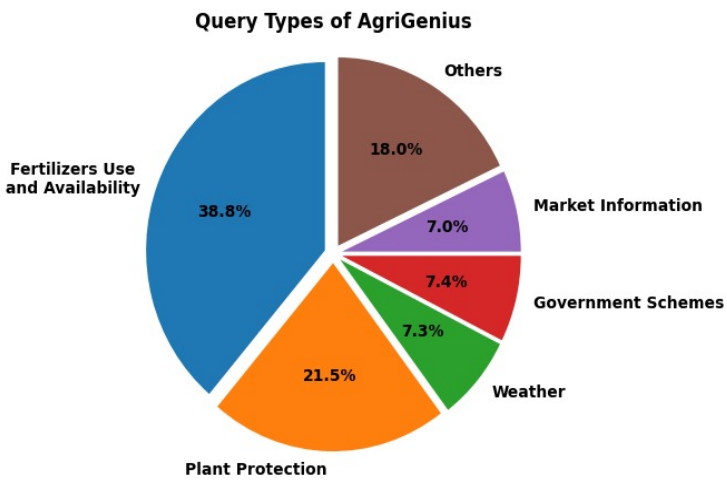


Fig. 3. Query Type Analysis

As shown in the Fig. 3, an analysis of user queries shows that 38.8% of interactions with the chatbot focus on "Fertilizers Use and Availability," highlighting farmers' concerns about nutrient management. Significant query categories also include "Plant Protection" (21.5%), "Weather" (7.3%), "Government Schemes" (7.4%), and "Market Information" (7.0%), reflecting the needs for a pest management, meteorological data, financial support, and market trends. The "Others" category, comprising 18.0% of queries, demonstrates the diverse information needs of the agricultural community.

TABLE I. QUERY TYPE REPORT

| Query Type | Percentage (%) |
|---|---|
| Fertilizers Use and Availability | 38.8% |
| Plant Protection | 21.5% |
| Others | 18.0% |
| Government Schemes | 7.4% |
| Weather | 7.3% |
| Market Information | 7.0% |

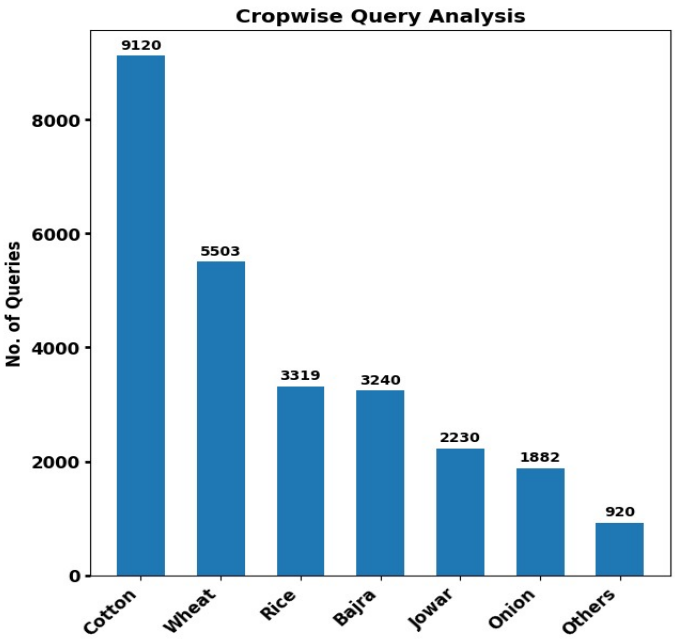


Fig. 4. Cropwise Query Analysis

As shown in the Fig.4, an analysis of crop-wise queries shows that the chatbot receives the most queries for cotton (9,120 queries), reflecting significant informational needs due to the crop's sensitivity to various factors. The chatbot provides the most accurate responses for cotton, ensuring precise and relevant advice. Wheat follows with 5,503 queries, this sub-category also having high response accuracy, which is vital in farming. Overall, the chatbot has a high level of accuracy to rice (3,319 queries) and bajra (3,240 queries) that provides quality information regarding cultivation. In jowar where there were 2,230 queries and in onion where there were 1,882 queries, the chatbot has further proved its efficiency in providing correct information on various crops. The same is true for less popular crops, and even if they are queried only 920 times,

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the accuracy of the chatbot does not deteriorate. This accuracy clearly demonstrates the relevance of giving detailed recommendations based on crop type and hence, increasing the usefulness of the chatbot in promoting better crop care and yield.

TABLE II. CROPWISE QUERY REPORT

| Crop | No. of Queries |
|---|---|
| Cotton | 9120 |
| Wheat | 5503 |
| Rice | 3319 |
| Bajra | 3240 |
| Jowar | 2230 |
| Onion | 1882 |
| Others | 920 |

2) **Metrics for Accuracy Evaluation**

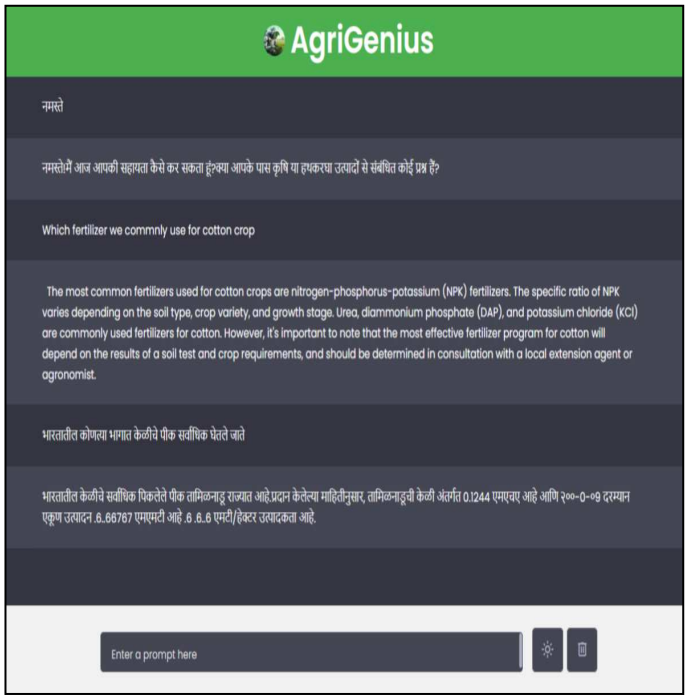


Fig. 5. User Interface of the Proposed System

As shown in the Fig.5, a crucial aspect of evaluating chatbot's effectiveness is assessing the accuracy of the information it provides to farmers. Here, we explore the metrics employed to gauge the chatbot's accuracy and the results obtained. Common metrics used to evaluate the accuracy of chatbots with information retrieval tasks include:

- **Precision:** This metric determines the proportion of the responses from Chatbot related to the user's query. It is defined as the number of relevant responses divided by the number of responses given by the chatbot.

$$Precision = \frac{TP}{TP + FP} \qquad (1)$$

- **Recall:** This metric specifies the proportion of relevant information within the knowledge base that the chatbot is able to retrieve in response to a user query. It is calculated as the percentage of correct responses to the total number of relevant responses possible.

$$Recall = \frac{TP}{TP + FN} \qquad (2)$$

- **F1-Score:** This metric provides a harmonic mean between precision and recall, offering a balanced view of retrieval accuracy.

$$F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall} \qquad (3)$$

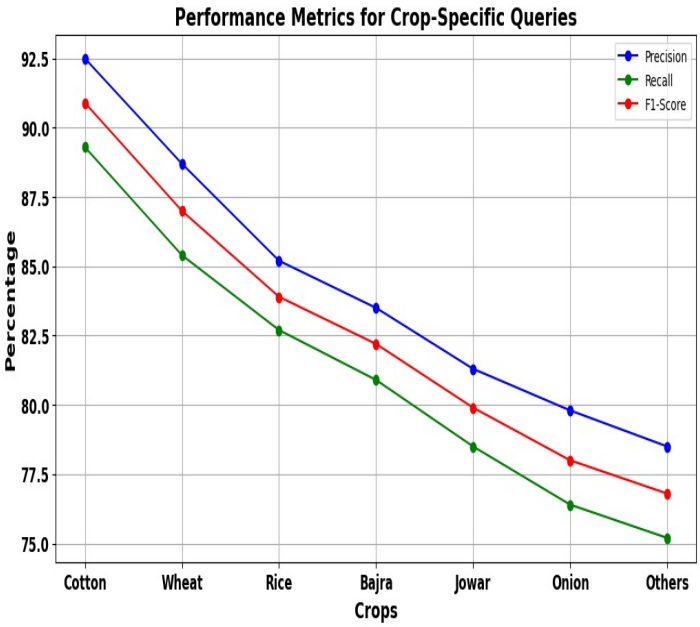


Fig. 6. Performance Metrics for Crop Queries

As shown in the above Fig.6, here the line graph effectively visualizes the overall performance of the chatbot regarding various Crops. In this case, several measures were considered in order to determine the effectiveness of the chatbot. For instance, if the chatbot processed 9,120 cotton queries and correctly identified 7,593 true positives, with 615 false positives and 911 false negatives, the precision, recall, the following results were obtained.

TABLE III. PERFORMANCE METRICS REPORT

| Crop | Precision | Recall | F1-Score |
|---|---|---|---|
| Cotton | 92.5 | 89.3 | 90.9 |
| Wheat | 88.7 | 85.4 | 87.0 |
| Rice | 85.2 | 82.7 | 83.9 |
| Bajra | 83.5 | 80.9 | 82.2 |
| Jowar | 81.3 | 78.5 | 79.9 |
| Onion | 79.8 | 76.4 | 78.0 |
| Others | 78.5 | 75.2 | 76.8 |

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### 3) Evaluation Results

We conducted an evaluation on a test set of farmer queries from all aspects of agriculture. Moreover, we evaluated the F1-Score, precision and recall. The evaluation outcomes show an efficiency of the RAG model and LLM methods in increasing the chatbot's accuracy. High precision implies that many of the responses that the chatbot provides are relevant to the user's query. High recall implies that chatbot is effective in identifying a large number of relevant data from the knowledge base. An F1- Score closer to one indicates better recall ability as well as precision and hence, better overall retrieval of results.

TABLE IV. CLASSIFICATION REPORT

| Parameters | Score |
|---|---|
| F1 Score | 97.90 |
| Precision | 98.50 |
| Recall | 96.00 |

## V. CONCLUSION

In Conclusion, the creation and utilization of an AI based chatbot that uses the RAG model mark a step, in providing agricultural support and sharing information. By adjusting and using language models the chatbot has shown its ability to offer accurate, contextually appropriate and timely advice to farmers effectively addressing their various needs and questions. The inclusion of an embedding database further boosts the chatbots performance by enabling access to information. However, there are still challenges to address like reducing bias ensuring use and optimizing for agricultural contexts. Areas that need more research and enhancement. Overall, it shows promise as a tool, for improving farm management practices boosting productivity and supporting agriculture growth. With improvements and fresh ideas, it could potentially transform how farmers obtain and apply knowledge.

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